

An xG Based Football Scouting System Using Machine Learning Techniques

Abhinav BV, Adhi Nandan J, Adithya S Gurikar, Aditya Joshi, Akshit Pandharkar and Prajwala T R

Department of Computer Science and Engineering, PES University, Bengaluru

Emails: adhinandanj@gmail.com, adithyagurikar10@gmail.com, aditya19joshi01@gmail.com, akshit.pan@gmail.com, abhibv04@gmail.com, prajwalatr@pes.edu

Abstract- In today's football landscape, scouting has evolved into a critical component for teams seeking hidden gems, established stars, and future talents. Amid this evolution, an innovative xG-based scouting system is poised to become a prominent fixture in the football lexicon. This system leverages the power of underlying stats like Expected Goals (xG) to meticulously evaluate players worldwide, recommending those who align with a team's unique needs and exhibit a significant potential.

Our approach uses a combination of various regression models and geometric distances to combine a large range of metrics and staggering statistics to function as a talent scout, ushering in a new era of player identification. It is able to identify players with huge potential and recommend them to clubs using it. This paper provides a comprehensive insight into the working of our pioneering xG-based scouting system, detailing its data-driven recommendations for teams. Ultimately this xG based scouting system enhances a team's ability to unearth the stars of tomorrow.

Keywords: Football Scouting, Football Player Performance, Machine Learning, Expected Goals, Regression, underlying statistics, Geometric Distance

I. INTRODUCTION

Scouting for players in football has always been dependent on the experience and perceptions of the scouts. They have been responsible for spotting players, uncovering abilities and unveiling in-depth tactics on the field. However with the advancements in technology and the availability of abundant data in football, there have been changes in the scouting techniques. Today's era is being driven by data and machine learning techniques, and have emerged as tools to improve the methodology of player scouting in general.

One crucial metric of player analysis in football is Expected Goals (xG). xG is a measure of the probability of a shot resulting in a goal by considering the factors such as distance from the goal, angle at which the shot is taken, number of defenders between the shooter and the goalpost, etc. By using metrics such as xG, we gain a complete understanding of the player and team performance beyond the traditional statistics like goals scored or assists made.

This paper introduces an innovative, cost-effective system for player scouting in football that combines xG based analytics with machine learning capabilities. By combining insights from xG and the advantage offered by machine learning algorithms, our system aims to revolutionize the search for players with hidden talents and skills in football. From a statistical viewpoint, our system provides an objective approach to scouting that enhances the

existing methodologies that are traditionally used by club scouts and analysts.

II. LITERATURE SURVEY

One of the Premier League's finest teams, Brentford FC, have adopted a 'moneyball approach' to footballing operations. Liam Callaghan Doyle et al. [5] looks into this approach and its feasibility. Brentford adopted a Moneyball model in the hopes of levelling the playing field. The results of the data analysis associated with Brentford's statistical model show that Brentford have grown in both playing and financial matters. Due to Brentford's analytical approach the club was able to outperform several richer clubs through smart acquisitions. This study shows that this approach is indeed a feasible one, with more unheralded players proving a major success for the club. Overall, underlying stats such as Expected goals and Expected Assists can be used to effectively scout or find players, who in the long run, have potential to reach their ceiling.

Predicting player performances using underlying stats such as Expected Goals and Expected Assists is the future of sports analytics, as depicted by Murtaza Saifi et al. [6], whose paper aims to determine how strong an attribute can underlying statistics bewhile predicting the player's performance. Fantasy sports has become an important application of such data. Ensemble based classification modelling such as Random Forest and Extreme Gradient Boosting have been used with and without the presence of underlying statistics and a 0.19% increase in accuracy 0.2% increase in F1 score is observed with the varying presence of data points.

Clustering of football players based on performance data can also be proven as an effective method. This is shown by Serhat Emre Akhanl et al. [7], where data is clustered based on a tailor-made dissimilarity measure. A suitable validation index is used, based on which the data is divided into two major clusters, one of which helps in the team's composition and the other can be used to find players very similar to the given profile of a player. This approach reduces information loss.

Football player performance analysis using Particle Swarm Optimization (PSO) is a commonly used method, depicted effectively by Ayanabha Jana et al. [8] Statistical data of various football players were analyzed and a correlation was established between their play style and quantifiable attributes. The next step after correlation would be to use PSO to simulate a match and compare two random players, having constrained their scores within the boundaries of the top recommended player, using the K-nearest neighbor algorithm. Lastly, a regression equation is

modelled using stepwise regression to estimate the net worth of the player based on their skill set, and predictions are performed using the optimal score obtained from PSO.

III. PROPOSED METHODOLOGY

The aim of the project is to use Machine Learning Techniques to find players with potential to improve drastically compared to their performance in the previous season. The crux of the project is focused on the underlying stats of the players namely xG (Expected Goals) and xA (Expected Assists). There is also a derived attribute called xGi (expected Goal Involvement) which is the sum of a players' xG and xA.

A. Data Collection and preprocessing

The dataset of the players was scraped from the website fbref.com. It contains key stats of more than 7000 players from the top eleven professional football leagues in the world including their underlying stats which are the focus in this project. The data collected is from the 2022-23 football season. The data was first manually cleaned to filter out stats irrelevant to certain playing positions. Players who had played less than 180 minutes during the course of the season were removed leaving about 6000 players in the database. It was required to create several additional attributes manually as well, such as 'Key Passes/90', 'Tackles/90', etc. This was necessary as the format of each league in the dataset is different from the others. So, simply having players' stats such as 'Key Passes' and 'Tackles' would not have been enough as the number of games played in a season can differ between the leagues putting players in smaller leagues at a disadvantage.

B. Defining Position based Parameters/ selecting target variables

Parameters were defined individually for each position on the pitch. The positions covered by the system are Centre-forward, Winger, Attacking Midfield, Central Midfield, Defensive Midfield, Wing-Back, Centre-back and Ballplaying Defender. Since each position satisfies different requirements in football, the parameters had to be defined individually for each position. It is also possible to select the qualities of a player from all the attributes provided using a Custom Selection feature in case none of the pre-defined playstyles meet the user's needs.

TABLE I. REGRESSION TEST RESULTS FOR CENTRE-FORWARDS

	MSE	RMSE	MAE	R ²
Linear Regression	0.00	0.00	0.00	0.99
Random Forest	0.00	0.01	0.00	0.98
XGBoost Regressor	0.00	0.01	0.00	0.96

C. Running regression tests

The parameters and the raw stats of the players are then entered as target variables into five different regression models: Linear Regression, Random Forest Regressor,

XGBoost Regressor, Lasso Regressor and Ridge Regressor. The raw stats of a player are the surface numbers we can see while watching the game and do not include underlying stats such as xG and xA. As mentioned before, different positions have different requirements and different regression models produce varying results for each position. For example, for Centre-forwards where the requirements are one-dimensional (Goals are the only requirement) Linear Regression tends to perform the best, whereas for Wing-Backs who have multiple roles (Defensive and Offensive requirements), the XGBoost or Random Forest models perform well. The models' output is evaluated using many parameters such as MSE, RMSE, MAE and R² values.

TABLE II. REGRESSION TEST RESULTS FOR BALL PLAYING DEFENDERS

	MSE	RMSE	MAE	R ²
Linear Regression	0.01	0.05	0.03	0.77
Random Forest	0.01	0.05	0.03	0.77
XGBoost Regressor	0.00	0.04	0.02	0.85

D. Using the best fit model

The model which produces the best output is chosen to find the best players in the selected position. The first set of regression models produce an output which contains the best players of the selected position in the 2022-23 season and also predicts which players will perform better in the upcoming seasons. The top players are determined by the regression models by assigning a weighted sum to each player based on their stats and predicted values. But these players have all performed well and have great stats which means they will be very expensive on the market. The aim of this scouting system is to find players who have gone under the radar, have the potential to perform well in the upcoming seasons and who can be signed on a small budget.

E. Incorporating xG using Geometric Distances

Next, the underlying stats of all players are loaded and the weighted sum values calculated by the regression models are mapped to each player in the underlying stats table. The top 10 players of the selected position are shortlisted and an average of their underlying stats are noted. This average is compared to all the other players in the database and players with the least Euclidean Distance and Manhattan Distance to the average are obtained. Finally, a weight is assigned to favorable xG and xA stats and the final output is obtained which is a combination of the geometric distances to the top players and favorable underlying stats.

IV. RESULTS AND CONCLUSIONS

The final output contains players whose underlying stats are similar to the top players in their respective positions and who have favorable xG and xA stats. This combination ensures that the player has the potential to perform well in the future. As on the surface, he might not appear as a top player but his underlying stats are similar to the best players

in his position. The reason he has not been able to convert his underlying stats into results can be due to various factors such as luck. When this player signs for the team using this system, it is probable that his luck will turn around and he will start matching his underlying stats with actual output on the pitch.



Fig. 1. Difference in players based on raw stats(top players) and underlying stats(recommended players)

As the figure above shows us, the model is clearly able to differentiate between players who have performed well in the past and found players who will perform well in the future. The table on the right contains players with favorable underlying stats who will be less expensive on the market.

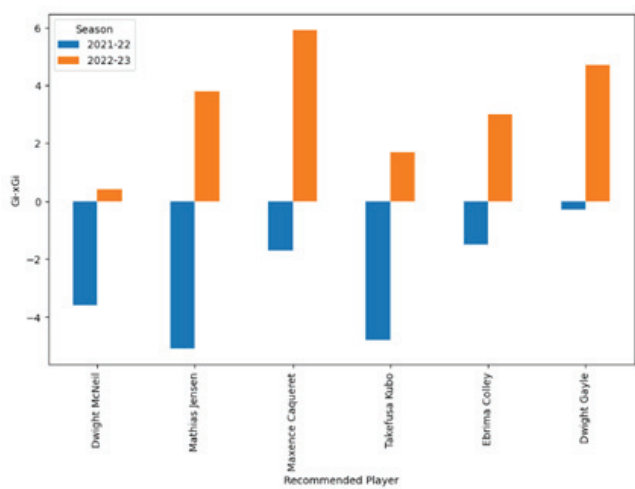


Fig. 2. Some players predicted to perform well by the model when the 2021-22 dataset was used

When the database of players from the 2021-22 season is given as input into the model, it manages to predict several players in different positions who went on to have great seasons in 2022-23. The results of our system also show how players lived up to their xG value in due course. Even after several great results from the model, it is not certain that every player predicted by it will have a great season the next time out. Ultimately, numbers and data are only meant to aid scouts in searching for players and finding things that humans cannot see with the naked eye. When signing a player for a football club, it is necessary to evaluate the player in all aspects such as health, manager's philosophy, playstyle and the player's commitment and love for the game which cannot be done through a computer.

It is recommended to start looking for players using the system, then evaluating each player recommended individually and thoroughly. Computers are great at reducing a list of 6000 potential candidates to about 30-40 by looking at all of their numbers but ultimately the decision to recruit a player should be up to the scouts with years of experience in the game. This system provides a great balance between computers and humans and ensures the best results are obtained through the combination of the two. This concept could also be expanded to cricket (Expected Runs(xR) and Expected Wickets(xW)) and rugby (Expected Points (xP)).

ACKNOWLEDGMENTS

We want to extend our appreciation to the individuals who have made contributions to this research on a xG based scouting system.

We are immensely grateful for the invaluable guidance and expertise provided by our mentors Dr. Prajwala T R and Dr. Sandesh B J.

A special thanks goes to Centre for Data Modelling Analytics and Visualization for their support and provision of resources.

We would also like to acknowledge our colleagues in the STARC community for their engaging discussions and helpful feedback.

REFERENCES

- [1] "FIFA 23 player dataset"- <https://www.kaggle.com/datasets/stefanoleone992/fifa-23-complete-player-dataset>
- [2] "Football statistics and history" - <https://fbref.com/en/players/>
- [3] "Big 5 European leagues stats"- <https://fbref.com/en/comps/Big5/Big-5-European-Leagues-Stats>
- [4] "xG stats for teams and players from the TOP European leagues"- <https://understat.com/>
- [5] "Is the Moneyball approach a feasible approach for success in Football (Soccer)? An analysis of Brentford's adoption of a statistical approach."-- Liam Callaghan Doyle (2022).
- [6] "Implementation of Machine Learning Techniques to Predict Player Performance using Underlying Statistics"- Murtaza Saifi
- [7] (2018)
- [8] "Clustering of football players based on performance data and aggregated clustering validity indexes" - Serhat Akhanli and Christian Hennig (2023)
- [9] "Football Player Performance Analysis using Particle Swarm Optimization and Player Value Calculation using Regression"- Ayanabha Jana and Sweetlin Hemalatha (2021).
- [10] "Some Issues in Distance Construction for Football Players Performance Data." -Akhanli, S. E., and C. Hennig. 2017, Archives of Data Science, Series A 2(1):17.
- [11] "Optimization modelling for analysing fantasy sport games"-Beli'en, J., Goossens, D. and Van Reeth, D. (2017). Infor 55(4): 275–294.
- [12] "Learning to predict soccer results from relational data with gradient boosted trees, Machine Learning"- Hub'a'cek, O., Sourek, G. and 'Zelezny', F. (2019). 108(1): 29–47.
- [13] "Mathematical programming as a tool for virtual soccer coaches: A case study of a fantasy sport game, International Transactions in Operational Research" -Bonomo, F., Dur'an, G. and Marenco, J. (2014),21(3): 399–414.
- [14] "Guest editorial: special issue on machine learning for soccer, Machine Learning"- Berrar, D., Lopes, P., Davis, J. and Dubitzky, W. (2019).108(1): 1–7.
- [15] "Sports Analytics for Football League Table and Player Performance Prediction"-Chazan-Pantazis V and Tjortjis C (2020), Proc. 11th Int. Conf. on Information, Intelligence, Systems and Applications (Piraeus) (IEEE) pp 1-8.